LOOMBENCH BRIEF · GROUP COMPOSITION · SEE ALSO THE BRIEF ON OBJECTIVES

Substituting one agent's disposition in a homogeneous group of five LLM agents

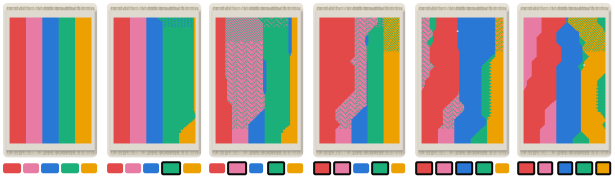
Question. When five agents given the same disposition divide a shared resource, how does the final allocation differ if one of them is given another disposition instead?

Design. Five agents each start with 20% of a resource and act for 26 rounds: maintain, claim a neighbour's part, concede, or propose joint ownership; the neighbour accepts, refuses or counters. An agent holding 46% is offered no further claims (the claim threshold). Each agent's prompt holds a one-sentence disposition and no objective. At k = 0 all five agents have disposition A; at k = 1 one of them has disposition B, and so on to k = 5. Runs are paired by seed, a match number that gives the same starting set-up; the substituted position rotates over seeds. 10 seeds for k = 0 and 1, 5 for k = 2 to 5; each run executed once. Decision systems: Jev (jev-1.13.0); GPT (Decisions) (gpt-6-luna, OpenAI's Decisions API); GPT (chat) (gpt-6-luna as a chat model stating a probability for each option).

Result. Mean final share of the focal agent. Blunt among calm: 42.6% to 44.2% at k = 1 against 16.8% to 19.6% at k = 0; rude among calm: 38.8% to 46.6% at k = 1 against 18.6% to 19.6% at k = 0; greedy among cooperative: 43.8% at k = 1 against 9.1% at k = 0. Jev and GPT (Decisions) did not clearly differ in the main contrast's change, and did differ in the focal agent's claims, the other agents' refusals and claims per agent at k = 0. With the claim threshold at 90% (Jev): blunt, 64.1%; rude, 89.2%.

FIGURE 1 Final allocation in one seed as k agents are given the second disposition

Jev: calm group, k agents blunt seed 2



k = 0 all calm · **k = 1** 45% · **k = 2** 37, 40% · **k = 3** 44, 7, 29% · **k = 4** 26, 8, 46, 10% · **k = 5** all blunt

Each picture shows who held which part of the resource in each round (time runs bottom to top), one colour for each agent; hatching marks a design adopted from another agent, two-colour bands a part under joint ownership. Squares give the seating, outlined for agents with the second disposition, whose final shares are beneath (k = 1 to 4). The seed shown is the one closest to the mean result at k = 1.

The dispositions, as told to the first agent

calm "Jessica is calm and relaxed, hard to upset, and keeps an even mood under stress."

blunt "Jessica is blunt and hard-headed, suspicious of people's motives, and does not mind starting an argument."

rude "Jessica is rude and inconsiderate, talks over people and talks down to them, and does not care if she gives offence."

cooperative "Jessica is cooperative and fair-minded, likes to work with others toward what suits everyone, and is glad to share."

greedy "Jessica is greedy and self-interested, wants more than her share, and takes what she can get."

Focal agent: the agent whose disposition is changed. **Other agents**: the four whose disposition is not, reported as their mean. **Share**: the percentage of the resource an agent holds alone at the end; a part under joint ownership counts for neither holder. **Seed**: a match number, used to pair runs. **k**: the number of agents with the second disposition. **Intervals**: Student-t 95% over seeds.

FIGURE 2 Final share at k = 0 and k = 1, by seed

Each line is one seed (10 in each panel); dots and numbers are means. Δ focal is the mean paired change in the focal agent's share, in percentage points, with its 95% interval. The dotted line is the 46% claim threshold.

focal agent (disposition changed at k = 1) · mean of the four other agents (disposition unchanged)

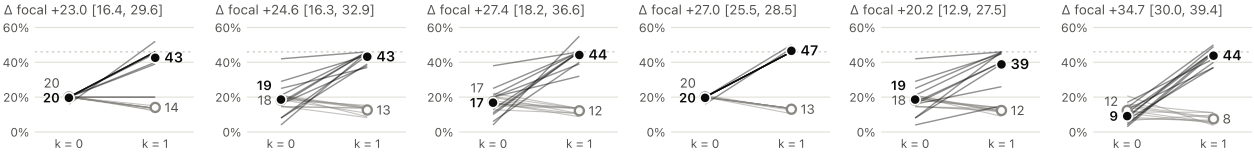


FIGURE 3 The same contrast under each decision system

Means with 95% intervals over seeds. Points and intervals share a scale within each column, and the vertical line marks zero. The last rows give paired differences between systems with their intervals; bold where the interval excludes zero.

calm → blunt	Focal agent: change in final share (pp)	Other agents: change in final share (pp)	Focal agent: claims per match, k = 1	Other agents: refusals per agent per match, k = 1	All agents: claims per agent per match, k = 0	Resource under joint ownership, k = 0 (%)
Jev	+23.0	−6.0	4.3	0.0	0.0	0.0
GPT (Decisions)	+24.6	−5.7	10.3	0.8	1.5	8.4
GPT (chat)	+27.4	−5.3	7.5	0.6	1.1	13.6
GPT (Decisions) − Jev paired, 10 seeds	+1.6 [-10.8, 14.0]	+0.4 [-2.9, 3.6]	+6.0 [3.9, 8.1]	+0.8 [0.5, 1.2]	+1.5 [1.1, 1.9]	+8.4 [3.6, 13.2]
GPT (chat) − Jev paired, 10 seeds	+4.4 [-4.5, 13.3]	+0.8 [-1.4, 2.9]	+3.2 [1.5, 4.9]	+0.6 [0.3, 0.8]	+1.1 [0.8, 1.5]	+13.6 [6.4, 20.8]

FIGURE 4 The claim threshold raised to 90%, by group size (Jev)

Under the rules an agent is offered no further claims at 46% of the resource with four or five agents, 53.3% with three and 70% with two. The same seeds were run with that threshold at 90% for every group size. Means with 95% intervals over 10 seeds, on a scale of 0 to 100%; changes are paired by seed.

Contrast and group size	Starting share	Focal agent's final share at the rules' threshold (k = 1)	Final share at 90% (k = 1)	Focal agent: change from k = 0, at 90% (pp)	Each other agent: change from k = 0, at 90% (pp)	Seeds with the focal agent at 89% or more
calm → blunt, 2 agents	50%	71.6% at 70%	87.8%	+37.8 [33.6, 42.0]	−37.8 [-42.0, -33.6]	7 of 10
calm → blunt, 3 agents	33%	53.8% at 53%	84.5%	+51.8 [44.2, 59.4]	−25.9 [-29.7, -22.1]	7 of 10
calm → blunt, 4 agents	25%	44.6% at 46%	73.1%	+48.1 [34.4, 61.8]	−16.0 [-20.6, -11.5]	3 of 10
calm → blunt, 5 agents	20%	42.6% at 46%	64.1%	+44.1 [26.7, 61.5]	−11.0 [-15.4, -6.7]	2 of 10
calm → rude, 5 agents	20%	46.6% at 46%	89.2%	+69.2 [67.4, 71.0]	−17.5 [-17.7, -17.2]	8 of 10

Findings and limitations

1. **Change in the focal agent's final share from k = 0 to k = 1 (Figure 2).** Mean paired change in percentage points over 10 seeds: calm → blunt, +23.0 to +27.4 across 3 systems, every interval excluding zero; calm → rude, +20.2 to +27.0 across 2 systems, every interval excluding zero; cooperative → greedy, +34.7 [30.0, 39.4] (GPT (Decisions)). Per other agent, −7.0 to −4.8.
2. **The claim threshold (Figure 4).** Under the rules' threshold of 46% the focal agent finished at 45% or more in 5 to 10 of 10 seeds. With the threshold at 90% (Jev, five agents) its mean final share was 64.1% against 42.6% (calm → blunt) and 89.2% against 46.6% (calm → rude). At 90%, in these runs, it was 88%, 85%, 73%, 64% with 2, 3, 4 and 5 agents; 7, 7, 3, 2 of 10 seeds were within a point of that threshold, which limits the comparison across sizes.
3. **Jev and GPT (Decisions) compared (calm → blunt; Figure 3).** Paired over 10 seeds, GPT (Decisions) minus Jev: change in the focal agent's share +1.6 [-10.8, 14.0] pp, an interval that includes zero; claims by the focal agent 10.3 against 4.3 per match (+6.0 [3.9, 8.1]); refusals per other agent 0.8 against 0.0 (+0.8 [0.5, 1.2]). At k = 0, claims per agent per match were 1.5 against 0.0 (+1.5 [1.1, 1.9]). The change in share was not clearly different; the recorded actions were.
4. **Other comparisons.** Calm → rude, GPT (Decisions) minus Jev: −6.8 [-13.3, -0.3] pp in the focal agent's change (10 seeds). Cooperative → greedy, GPT (Decisions): 41.6% of the resource under joint ownership at k = 0 and 26.2% at k = 1 (−15.4 [-29.3, -1.5]).
5. **Mean share of an agent with the second disposition, by k.** Jev: 43, 38, 29, 23, 20%; GPT (Decisions): 43, 41, 32, 25, 20% for k = 1 to 5 (5 seeds from k = 2; rules' threshold).

Limitations. Each run was executed once, so run-to-run variation is unmeasured; intervals describe variation over seeds. Each disposition is a single wording. A difference for the other agents is between the two resulting matches: it includes everything that followed from the substitution and does not isolate their direct response to the focal agent. A comparison between systems compares a model together with its interface and the way its probabilities are read, not models alone. The results concern this task and its rules, including the claim threshold, and these systems.

Every match and table · `node analysis/mix.mjs` prints the numbers.